THE QUEST

To prove CTI and the well:

phase - 0.

derivation of PTE:

$$i\hbar \frac{\partial |\Psi(t)\rangle}{\partial t} = \hat{H} |\Psi(t)\rangle$$ (Schrödinger time dependent)

↳ sol^n:

$$|\Psi(t)\rangle = \hat{U}(t) |\Psi(0)\rangle$$

$$U(t) = e^{\frac{-iHt}{\hbar}} \rightsquigarrow e^{-iHt}$$

for one $\hat{H}$

but if $\hat{H} = \sum_{i=1}^{N} \hat{H}_i$

and Trotterised:

$$U(t) = \prod_{n=1}^{N} \left(\prod_{m=1}^{M} e^{-iH_m \delta_n} \right) \quad \delta = \frac{t}{n}$$

now for a single layer, it forms a weighted sum of operators, so for layer k: $\rho \in [0,1]$ $\tau \in [0, \tau_{max}]$

$$B_k(\rho, \tau) = \sum_{j}^{M} \rho_{j,k} \left(e^{-iH_j \tau_{j,k} \delta} \right)$$

$$A = \prod_{k=1}^{n} B_k(\rho; \tau)$$

↳ probably non-unitary ($A^\dagger A \neq \mathbb{I}$)

so polar decomposition would be used

Polar Decomposition:

taking $-1/2$ power of a matrix projects it to be orthonormal basis:

if you have matrix A:

construct $S \Rightarrow A^\dagger A$; $S^{-1/2} = (A^\dagger A)^{-1/2}$

now $A = QP$, Q is unitary $Q^\dagger Q = \mathbb{I}$

P is hermitian $P^\dagger = P$

now $A^\dagger A = (QP)^\dagger (QP) = P^\dagger \cancel{Q^\dagger Q} P$

$\Rightarrow P^2 = (A^\dagger A)$ ($Q^\dagger Q = \mathbb{I}$)

$\therefore (A^\dagger A)^{-1/2} = P^{-1}$ ($P^{-1} P = \mathbb{I}$)

now $A(A^\dagger A)^{-1/2} = Q \cancel{P P^{-1}}$

so $\boxed{A(A^\dagger A)^{-1/2} = Q}$ $\Rightarrow$ purely the rotation part of A

using this in PTE:

$$B_k = \sum_j^N \rho_{j,k} e^{-i H_j S_{j,k}}$$

$$U_k = (B_k)_{\text{Polar}} = B_k (B_k^\dagger B_k)^{-1/2}$$

$$\hat{U}_{\text{safe}} = \prod_k^N U_k$$

↳ global PTE

The Structural well:

Phase - ①

→ fast descent in early iterations, stabilizes later.

→ need to theoretically prove this:

the polar projection shows that the new unitary does not depend on stretching whether through noise or a field.

only depends on rotation

→ most noise stretches. if it can't affect U, then U goes down the landscape and settles at the bottom ⇒ THE WELL

this is the intuition.

→ stretching means going out of the unitary manifold (most noise types do that)

Polar projection form:

$B = QP$

$B + \delta B = (Q + \delta Q)(P + \delta P)$

$B + \delta B = QP + Q\delta P + \delta Q P + \cancel{\delta Q \delta P}$ ($\delta Q \to 0$)

so

$\delta B = \delta Q P + Q \delta P$

now: $P = P^\dagger$

so $\delta P = \delta P^\dagger$

and $Q^\dagger Q = \mathbb{I}$

so by product rule

$$Q^\dagger \delta Q + \delta Q^\dagger Q = 0$$

let $Q^\dagger \delta Q = \Omega$; $(Q^\dagger \delta Q)^\dagger = \delta Q^\dagger Q$

$$\Omega^\dagger = \delta Q^\dagger Q$$

$$\therefore \Omega + \Omega^\dagger = 0$$

on the polar projection form:

$$\delta B = \delta Q \cdot P + Q \cdot \delta P$$

$$Q^\dagger \delta B = (Q^\dagger \delta Q) P + (Q^\dagger Q) \delta P \quad (Q^\dagger Q = \mathbb{I})$$

$$Q^\dagger \delta B = \Omega P + \delta P$$

so $\delta P = Q^\dagger \delta B - \Omega P$ — (1)

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or ① take full transpose

$$\delta P^\dagger = (Q^\dagger \delta B)^\dagger - (\Omega P)^\dagger$$

$$\delta P = \delta B^\dagger Q - P \Omega^\dagger \quad (2)$$

$\hookrightarrow + P\Omega$

① = ②

$$\therefore \delta B^\dagger Q + P\Omega = Q^\dagger \delta B - \Omega P$$

$$\rightarrow \underline{\Omega P + P\Omega = Q^\dagger \delta B - \delta B^\dagger Q}$$

↓ sylvester equation solved by computer

↳ you get Ω

now $\Omega = Q^\dagger \delta Q$

$$Q\Omega = (QQ^\dagger)\delta Q \quad (QQ^\dagger = \mathbb{I})$$

$$\boxed{Q\Omega = \delta Q}$$

BUT NOW

if $B^\dagger B = \mathbb{I}$, $B = QP$

$$(QP)^\dagger (QP) = \mathbb{I}$$

$$P^\dagger Q^\dagger Q P = \mathbb{I} \quad (Q^\dagger Q = \mathbb{I})$$

$$P^\dagger P = \mathbb{I}, \text{ but } P = P^\dagger$$

so $P^2 = \mathbb{I}$, $\boxed{P = \mathbb{I}}$, $B = Q$

plug in sylvester equation:

$$\Omega \mathbb{I} + \mathbb{I} \Omega = Q^\dagger \delta B - \delta B^\dagger Q$$

$$2\Omega = Q^\dagger \delta B - \delta B^\dagger Q$$

$$\Omega = \frac{1}{2}(Q^\dagger \delta B - \delta B^\dagger Q)$$

now $\delta Q = B\Omega = B\left(\frac{1}{2}(B^\dagger \delta B - \delta B^\dagger B)\right)$

$$\boxed{\delta Q = \frac{1}{2}(\delta B - B \delta B^\dagger B)}$$

if δB is purely stretching:

so $\delta B = Q \delta P$ (polar projection)

$$\delta B = B \delta P$$

$$\delta Q = \frac{1}{2}(B \delta P - B (B \delta P)^\dagger B)$$

$(BSP)^\dagger = SP^\dagger B^\dagger = SPB^\dagger \quad \because S^\dagger = S$

So $\delta Q = \frac{1}{2}(BSP - BSP(B^\dagger B))$ (II)

$$\boxed{\delta Q = \frac{1}{2}(BSP - BSP) = 0} \quad (I)$$

⇒ if the change is purely stretching causes no change in state

⇒ NOISE MOSTLY TRIES STRETCHING OUTSIDE THE MANIFOLD IN SU(N)

But this is a local definition

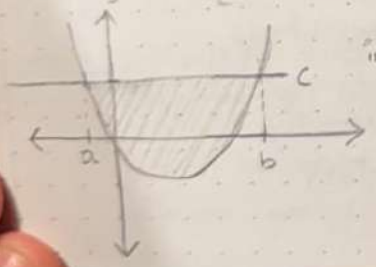
we need to generalise this to the entire structure.

Sublevel Sets:

for $f(x)$ a function (2-D), the set on $f(x), L$ is a sublevel set if it has all x in domain that give $y = f(x)$ less or equal to a constant:

$f: A \to B, \quad c = 5$

$L_5(f) = \{x \in A \mid f(x) \leq 5\}$



in 2-D it is that shape and $L_c(f) = [a, b]$

for 3-D, $f: \mathbb{R}^2 \to \mathbb{R}$, $L_c(f) \triangleq \{(x,y) \in \mathbb{R}^2 \mid f \leq c\}$

for TOPAZ, the function is

$$L(\beta, T) = 1 - |\langle \Psi_{target} | U_{safe}(\beta, T) | \Psi_0 \rangle|^2$$

better $U_{safe}(\beta, T) \to L(\beta, T)$ smaller, at $L = 0$ is perfect.

here sublevel set $W_c(L) = \{(\beta, T) \mid L(\beta, T) \leq c\}$

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in 3D func. we have a derivative vector that points to the highest increase direction:

for $f(x,y) \Rightarrow \nabla f = \begin{bmatrix} \frac{\partial f}{\partial x} \\ \frac{\partial f}{\partial y} \end{bmatrix}$, called gradient

→ critical points are points where $\nabla f = 0$

→ so, local extrema, inflection and saddle points and global extrema

→ for a single minimum, and for a defined sublevel constant c, a Basin of Attraction $B_c(x^*)$:

$B_c(x^*) \subseteq L_c(f)$ (sublevel set) → sublevel containment

↳ (Basin is in sublevel set)

if $x \in B_c(x^*)$, $\nabla f(x) = 0$ then $\boxed{x = x^*}$

↳ (only minimum, single minimum in a basin) (2)

$-\nabla f(x) \cdot \vec{n}(x) < 0 \quad \forall x \in \partial B_c(x^*)$ (↳ boundary of basin)

↳ (all points ∇f always point out)

now: for $f: A \to B$

$M = \{x_k^* |_{k=1}^{n} \mid \nabla f(x_k^*) = 0 \mid \nabla^2 f(x_k^*) > 0\}$ → set of all minima

so $W = \{B(x_i^*) \mid x_i^* \in M\}$ → set of all basins

where

$B(x_i^*) \cap B(x_j^*) = \phi \quad \forall i \neq j$ → all basins disjoint

$\bigcup_{i=1}^{k} B(x_i^*) = A \setminus \{\}$ ③

↳ all maxima/saddle/infection points

↳ union of all basins = domain except other points

For TOPAZ:

$f = L(P, T)$

$\mathcal{X} = [0,1]^M \times [0, T_{max}]^M$; N = no. entangling pairs

$L: \mathcal{X} \to \mathbb{R}$

$L(P,T) = 1 - |\langle \Psi_{ideal} | \hat{U}_{safe} | \Psi_0 \rangle|^2$

Gradient Invariance:

① Differential of pure quantum fidelity:

let $\hat{U}_{safe} |\Psi_0\rangle = |\phi\rangle$

$F = |\langle \Psi_{ideal} | \phi \rangle|^2 = \langle \Psi_{ideal} | \phi \rangle \langle \phi | \Psi_{ideal} \rangle$

$\delta|\phi\rangle = (\delta U_{safe}) |\Psi_0\rangle$ — ①

↳ $\delta\langle\phi| = \langle\Psi_0| (\delta U_{safe}^{\dagger})$ — ②

$L = 1 - F$

$L = 1 - \langle \Psi_{id} | \phi \rangle \langle \phi | \Psi_{id} \rangle$

$\delta L = 0 - (\langle \Psi_{ideal} | \phi \rangle \cdot \delta \langle \phi | \Psi_{id} \rangle + \delta \langle \Psi_{id} | \phi \rangle \cdot \langle \phi | \Psi_{id} \rangle)$

$\delta L = -(\langle \Psi_{id} | \delta\phi \rangle \langle \phi | \Psi_{id} \rangle + \langle \Psi_{id} | \phi \rangle \langle \delta\phi | \Psi_{ideal} \rangle)$

from ① & ②:

$\delta L = -(\langle \Psi_{id} | \delta U_{safe} | \Psi_0 \rangle \langle \phi | \Psi \rangle + \langle \Psi | \phi \rangle \langle \Psi_0 | \delta U_{safe}^{\dagger} | \Psi \rangle)$

$= -(\langle \phi | \Psi \rangle \langle \Psi | \delta U_{safe} | \Psi_0 \rangle + \langle \Psi | \phi \rangle \langle \Psi_0 | \delta U_{safe}^{\dagger} | \Psi \rangle)$

$(\langle \Psi | \delta U_{safe} | \Psi_0 \rangle)^{\dagger} = \langle \Psi_0 | \delta U_{safe}^{\dagger} | \Psi \rangle$

and if $z = a + ib$, $z + z^* = 2(\text{real}) z$

$$\boxed{\delta L = -2\,\text{Re}(\langle \phi | \Psi \rangle \langle \Psi | \delta U_{safe} | \Psi_0 \rangle)}$$

$\delta U_{safe} = \nabla P_v [\delta B]$

and because for noise, H is hermitean only)

$\delta B = 0$ (we proved earlier)

so $\boxed{\nabla P_v [0] = 0}$

so $\boxed{\delta L = 0}$ for hermitean change caused by noise

↳ gradients are invariant ④

$\nabla L_{noisy} = \nabla L_{ideal} + \nabla V_{noise}$

$\nabla V_{noise} = 0$

$\nabla L_{noisy} = \nabla L_{ideal}$

if $\forall$ points $\nabla L_{noisy} = \nabla L_{ideal}$

$\because -\nabla L_{ideal} = -\nabla L_{noisy}$

(in reality $-\nabla L_{ideal} \approx -\nabla L_{noisy}$)

the only thing that can be different is that $\nabla L_{ideal} + C = \nabla L_{noisy}$

$\because J(\nabla L_{ideal}) = J(\nabla L_{noisy})$

so the map is differentiable and therefore smooth

$\therefore \nabla L_{noisy}$ is approximately diffeomorphic to ∇L_{ideal}

↳ diffeomorphic invariance

A homeomorphism needs to be bijective, continuous and invertible, Diffeomorphs are all that plus differentiable.

homeom-
Diffeom-

EVERY DIFFEOMORPH IS A HOMEOMORPH

so all topological properties are invariant ⑤

×

Frobenius Norm:

sum of squares of all objects in the matrix !

F^2 = sum of squares

F of a matrix $\Rightarrow A = \|A\|_F$

$F^2 = \|A\|_F^2 = Tr(A^\dagger A)$

matrix here is $B_k^\dagger B_k - \mathbb{I}$

$\|B_k^\dagger B_k - \mathbb{I}\|_F^2 = Tr\left(\left(B_k^\dagger B_k - \mathbb{I}\right)^2\right)$

$Tr\left((B^\dagger B)^2 - 2B^\dagger B + \mathbb{I}\right)$

$= Tr(B^\dagger B)^2 + Tr(-2B^\dagger B) + Tr(\mathbb{I})$

$4B B^\dagger B \quad \circ \quad -2(2B) \quad + 0$

$= 4B(B^\dagger B - \mathbb{I})$

↳ if $B^\dagger B \neq \mathbb{I}$, this has a value

$\nabla_B (B^\dagger B)(B^\dagger B)$

$\Rightarrow (BB^\dagger B) + (BB^\dagger B)$

$(BB^\dagger B)$

$(BB^\dagger B)$

$= 4BB^\dagger B$

in a regularizer that penalizes terms that only stretch,

$$L = L_{init} + \lambda \sum_{k=1}^{r} \|B^T B - I\|_F^2 - f(B)$$

so $\nabla_{B_k} f(B) = 4B(B^T B - I)$

and it is non-zero. It feels a push.

↳ if B_k is purely stretching $f(B)$ gets larger, so it eventually tries pruning that B_k out.

↓

WHY IT PRUNES OUT

⑥

with ① - ⑥ you get the full framework and proof of the structural well.

To prove the existence of the well:

$\mathcal{X} = [0,1]^M \times [0, \tau_{max}]^M$

↳ this is compact (closed and bounded)

EVT (Extreme value Theorem):

if a function that is continuous is defined on a compact space it will have global extrema

$L(\rho, \tau)$ is continuous and defined on the compact parameter space $\mathcal{X}$.

↳ IT HAS A GLOBAL MINIMUM

→ Entanglement causes non-linearity of $L(\rho, \tau)$

↳ this by non-commutativity, causes more minima locally and forces the loss landscape to become non-convex.

→ it is SUFFICIENT, NOT STRICTLY NECESSARY

But more entanglement ⇒ more wells

→ The observed family across different random seeds is a consequence of the pruning mechanism

"WELL" PROVED BRUH

To sketch a graph → need $f'(x)$ and $f''(x)$

-ve → decreasing

+ve → up

a, c, b

put values from intervals

→ $f'(x)$ ⇒ dec → inc (min) ⇒ inc → dec (max)

↳ a and b from $f'(x) = 0$

→ $f''(x)$ ⇒ up/down → flips → inflection point

for $f'(x)$ → find fro... and $f''(x)$

Phase - II

The Equivalence Gap to be addressed:

$|L_{ideal} - L_{hardware}| \leq K \cdot D$

↳ state drift / trace distance

D is hard to calculate

but Fuchs-Van de Graaf saves us:

↳ they asserted that

$D \leq \sqrt{1-F}$

so you get

$|L_{ideal} - L_{hardware}| \leq K\sqrt{1-F}$

if there is a small error ε

$F \approx 1 - c \cdot \varepsilon^2$

$|L_{ideal} - L_{hardware}| \leq K\sqrt{1-(1-c\varepsilon^2)}$

↳ $|L_{ideal} - L_{hardware}| \leq K\sqrt{c}\,\varepsilon$

↳ its linear so if ε is small enough the well transfers from the ideal landscape to the real landscape

Equivalence GAP Addressed!

Phase - III

Lie Algebra to show LPTE beats HEA:

HEA is extremely expressive ⇒ too expressive → too many barren plateaus

↳ the DLA reaches close to $SU(2^N)$

LPTE doesn't increase DLA dimension if isolated

↳ $[X_1X_2, Y_1Y_2] = 0$ ↳ $\{\sum X_iX_{i+1}, Y_iY_{i+1}, Z_iZ_{i+1}\}$

these form an abelian subalgebra

- a single LPTE block is highly constrained, predictable and doesn't explore the full Hilbert space

BUT

LPTE Chain: $[X_1X_2, Y_2X_3]$ ⇒ new operator

→ when you chain overlapping pairs, the commutators generate new non-local directions, causes DLA to grow, expanding the reachable manifold but strategically and preserves symmetries

⇒ This allows TOPAZ to prune out abelian local blocks completely with changing the reachable manifold

LPTE BEATS HEA

Phase-IV

Why τ cluster around $\tau \approx \pi/2$

EVEN THOUGH

standard QM says max entangling power comes at $\tau = \pi/4$:

$$B(\tau) = \left(e^{i\tau XX} \cdot e^{i\tau YY} \cdot e^{-i\tau ZZ}\right)$$

$$e^{-ixA} = \cos(x)\,\mathbb{I} - i\sin(x)\,A$$

at $x = \pi/2$

$$e^{-i\pi/2\,XX} = \cos(\pi/2)\,\mathbb{I} - i\sin(\pi/2)\,XX = -i\,XX$$

so others are

$$B(\tau) = -i\,(XX + YY + ZZ)$$

$$P_U(B) = B\,(B^\dagger B)^{-1/2}$$

$XX + YY + ZZ = 2\,SWAP - \mathbb{I}$

$$B(\tau) = -i\,(2\,SWAP - \mathbb{I})$$

$$B^\dagger = i\,(2\,(SWAP)^\dagger - \mathbb{I}) = -B$$

$$(-B)^{2\cdot -1/2} = \left((2\,SWAP - \mathbb{I})^{2}\right)^{-1/2} = (M^2)^{-1/2}$$

Eigenvalues:

Symmetric $\Rightarrow$ SWAP has $+1$

M has $2(1) - 1 = +1$

antisymmetric $\Rightarrow$ SWAP has -1

M has $2(-1) - 1 = -3$

M^2 would have symmetric $= +1$.

antisymm $= +9$

$(M^2)^{-1/2}$ would have $1^{-1/2} = +1$ symmetric

$9^{-1/2} = \left(\frac{1}{9}\right)^{1/2} = +\frac{1}{3}$ antisymme

$i\,M(M^2)^{-1/2}$ would have:

$-i\,(+1)\,(+1) \Rightarrow -i$ for symmetric

$-i\,(-3)\,(+1/3) \Rightarrow +i$ for antisymmetric

A MATRIX THAT IS $+1$ ON SYMMETRIC AND -1 on ANTISYMMETRIC IS A SWAP gate

so gate with $-i$ on sym. and $+i$ on Anti. is $\boxed{-i\,SWAP}$

↳ This a maximally entangling gate.

↳ It is at $\boxed{\tau = \pi/2}$

$f(x)$ dec	inc
$f'(x)$ −ve	+ve

$f(x)$ concave down	concave up
$f'(x)$ dec	inc
$f''(x)$ −ve	+ve

For rates, put down variables: if they change each other, → use similar triangles, or given ratios

↳ CONES!

orginal $C(x)$, $P(x)$, $R(x)$ → $C'(x)$, $P'(x)$, $R'(x)$

$P(x) = R(x) - C(x)$

$P'(x) = R'(x) - C'(x)$

$f'(x) = 0$ gives a, b as crit points

Phase V:

Noise Model transferability:

noise = $\mathcal{H} = (\gamma_{depol}, \gamma_{amp}, \lambda_{phase})$

TOPAZ:

Params $\Rightarrow \chi = (\beta, \tau)$

Loss function $\Rightarrow \mathcal{L} = L(\alpha, \mathcal{U})$

$\mathcal{U} \rightarrow$ operator

$\mathcal{U} + \delta\mathcal{U} \Rightarrow$ change from noise

$\delta\mathcal{U} = [(\mathcal{Q} + \delta\mathcal{Q}) \cdot (P + \delta P)] \rightarrow$ all noise

$\mathcal{Q} \approx 0, \delta\mathcal{Q} \approx 0$

$\therefore \delta\mathcal{U} = P + \delta P$

$\delta\mathcal{U}$ depends on $\mathcal{H}$

if you change vector $\mathcal{H}$, $\delta\mathcal{U} = P + \delta P$

($\delta\mathcal{U}$ is still almost purely stretching)

so $|L(\alpha, \mathcal{H}_1) - L(\alpha, \mathcal{H}_2)| \leq \kappa \|\mathcal{H}_1 - \mathcal{H}_2\|$

due to projection $\kappa \rightarrow 0$

so the difference is almost 0

↳ the well is Lipschitz continuous (and α invariant) to noise parameters

Phase VI

The 4 properties:

1. compact: if $\mathcal{X} = [0,1]^M \times [0, t_{max}]^M$ is the domain space, or the domain space is closed and bounded

2. continuous: $L(\mathcal{U})$ is Lipschitz continuous by fidelity being Lipschitz continuous on the active manifold

3. connectedness: parameter domain $\mathcal{X} = [0,1]^M \times [0,T]^M$ in Topology a closed interval $[a, b]$ is convex cartesian product of convex sets is convex $\therefore \mathcal{X}$ is convex set.

Topology states that a convex set is path connected

Well is a connected component of the sublevel set on $\mathcal{X}$ $\therefore$ well (Wc) guarantees path-connected-ness

4. Convexity: $L_{primary}$ is non-convex

But the regulariser is in the form

$\lambda \Sigma_k \|B_k^\dagger B_k - \mathbb{I}\|_F^2 \Rightarrow \lambda \|X\|_F^2 = f(X)$

$\|X\|_F^2 = \mathrm{Tr}(X^T X)$ is a strictly convex quadratic form

$\nabla_x^2 f(x) \Rightarrow$ positive definite

↳ $\nabla_x^2 L_{Total} = \nabla_x^2 L_{prim} + \lambda \nabla_x^2 f$

as long as λ is large enough

$\nabla_x^2 f$ can force its curvature onto L_{prim}, even if it is not positive definite

↳ any minimum in L_{Total} is strictly a local minimum

↓

By Tikhonov Regularisation Theory

DONE